A PREDICTIVE SYSTEM FOR IDENTIFYING BOT - GENERATED AD CLICKS

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report **"A PREDICTIVE SYSTEM FOR IDENTIFYING BOT - GENERATED AD CLICKS"** is the bonafide work of

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who carried out the project work under my supervision.

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ABSTRACT

The Ad Click Fraud Detection Dashboard is an advanced system developed to identify and mitigate fraudulent activity in online advertising. Click fraud is a pervasive issue in digital marketing, where automated bots or malicious actors generate false clicks on advertisements. This not only leads to substantial financial losses for advertisers but also skews the analytics used to measure campaign effectiveness.

This project leverages machine learning models, specifically RandomForest and XGBoost classifiers, to predict whether an ad click originates from a genuine human user or a fraudulent source. The system is trained on historical click data with features such as IP address, application ID, device type, operating system, channel, and click timestamp. By analyzing these features, the models can accurately distinguish between human and bot clicks based on patterns and anomalies in user behavior.

The prediction models are integrated into a user-friendly web dashboard built with Flask, HTML, CSS, and JavaScript. The dashboard provides an interactive interface where users can input click-related data and obtain real-time predictions.

The primary objective of this system is to empower advertisers and businesses to detect suspicious clicks proactively, optimize their advertising budgets, and improve the accuracy of marketing analytics. This project demonstrates the practical application of machine learning for cybersecurity in digital advertising and provides a scalable foundation for future enhancements, such as real-time monitoring and integration with larger ad networks.

TABLE OF CONTENT

LIST OF FIGURES

LIST OF ABBREVATIONS

ABBREVATIONS	DESCRIPTIONS
SVM	Support Vector Machine
DT	Decision Tree
RF	Random Forest
XGB	XGBoost
ML	Machine Learning
AI	Artificial Intelligence
DL	Deep Learning